

Impact Statement

Optimizing RAG Document Retrieval for Agronomic Advice

Team Name	RasDashen
Selected Project	Optimizing RAG Document Retrieval for Agronomic Advice
Challenge	Challenge 3 — Impact Statement
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Purpose

Smallholder farmers frequently need fast, trustworthy answers to questions about crop disease, pests, soil health, and climate adaptation, but agricultural extension officers are stretched thin and the written extension material that exists is scattered across many documents. Our project builds a retrieval system that, given a farmer's question, returns the most relevant extension documents from a knowledge base — using hybrid BM25 and dense-embedding search rather than a generative model, so every answer traces back to a real, verifiable source.

Positive Impacts

The most direct beneficiaries are smallholder farmers and the extension workers who support them: faster access to the right source document means fewer delays acting on a crop problem, which can directly affect yield and household income. Extension organisations also benefit operationally, since a retrieval layer can help officers triage which reference material to consult first rather than searching manually. Because outputs are grounded in real documents rather than generated text, this benefit comes with lower risk of confidently wrong advice compared to a purely generative system.

Potential Risks and Harms

The main risk is uneven document coverage: publicly available extension material is denser for major export crops and staples than for regionally important but less documented crops, and skews toward English. A farmer asking about an under-covered crop or in a local language could receive a lower-quality or irrelevant top-5 result, potentially worse than no tool at all if trusted uncritically. There is also a technical risk of retrieval errors surfacing an outdated or context-mismatched document with confident framing, and an economic risk that any future deployment could bypass rather than support existing extension officers.

Values

The design promotes transparency (every result is a real, checkable document), fairness (an explicit commitment to auditing crop/language coverage rather than accepting whatever is easiest to source), and community well-being (positioning the tool to support, not replace, extension officers).

Mitigation Strategies

We audit topical and language coverage of the knowledge base and document known gaps rather than presenting the system as comprehensive; we keep the system retrieval-only so outputs stay traceable; and we frame the tool as a decision-support aid for extension workers and farmers, not a replacement for professional

agronomic judgement.

Limitations and Uncertainties

Our nDCG@5 evaluation reflects performance on the labelled queries available to us, which may not represent the full diversity of real farmer questions, and our assessment of coverage gaps is based on manual review rather than a systematic audit, so some gaps may remain undetected at this stage.

Prepared for Challenge 3 (Impact Statement) submission. Fields in brackets should be completed with team-specific details before submission.